SEC2_APM1110-FA4-DABU, L.

GitHub: https://github.com/dabulira20/Prob-ProbDi-FAs/tree/main/FA4

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1. Exercise 2

A binary communication channel carries data as one of two sets of signals denoted by 0 and 1. Owing to noise, a transmitted 0 is sometimes received as a 1, and a transmitted 1 is sometimes received as a 0. A transmitted 0 is correctly received with probability 0.95, and a transmitted 1 is correctly received with probability 0.75. Also, 70% of all messages are transmitted as a 0.

Notation

- T_0: a 0 is transmitted, T_1: a 1 is transmitted
- R_0: a 0 is received, R_1: a 1 is received

Given probabilities

- $P(T_0) = 0.70$, therefore $P(T_1) = 0.30$
- $P(R_0 \mid T_0) = 0.95$, therefore $P(R_1 \mid T_0) = 0.05$
- $P(R_1 \mid T_1) = 0.75$, therefore $P(R_0 \mid T_1) = 0.25$

```r
P_T0 <- 0.70
P_T1 <- 0.30

P_R1_given_T0 <- 0.05   # 0 transmitted, but received as 1
P_R1_given_T1 <- 0.75   # 1 transmitted, correctly received as 1
```

(a) Probability that a 1 was received

Using the Law of Total Probability:

$$P(R_1) = P(R_1 \mid T_0)\, P(T_0) + P(R_1 \mid T_1)\, P(T_1)$$

```r
P_R1 <- P_R1_given_T0 * P_T0 + P_R1_given_T1 * P_T1
P_R1
```

```
## [1] 0.26
```

Answer: $P(R_1) = 0.26$, so there is a **26% probability** that a 1 was received.

(b) Probability that a 1 was transmitted, given that a 1 was received

Using Bayes' Theorem:

$$P(T_1 \mid R_1) = \frac{P(R_1 \mid T_1)\,P(T_1)}{P(R_1)}$$

```r
P_T1_given_R1 <- (P_R1_given_T1 * P_T1) / P_R1
P_T1_given_R1
```

```
## [1] 0.8653846
```

Answer: $P(T_1 \mid R_1) \approx 0.8654$, so there is roughly an **86.54% probability** that a 1 was transmitted given that a 1 was received.

Exercise 7

There are three employees at an IT company: Jane, Amy, and Ava, who do 10%, 30%, and 60% of the programming, respectively. 8% of Jane's work, 5% of Amy's work, and 1% of Ava's work contains errors.

Notation

- J, A, V: the program was written by Jane, Amy, or Ava respectively
- E: the program contains an error

Given probabilities

- $P(J) = 0.10$, $P(A) = 0.30$, $P(V) = 0.60$
- $P(E \mid J) = 0.08$, $P(E \mid A) = 0.05$, $P(E \mid V) = 0.01$

```r
P_J <- 0.10
P_A <- 0.30
P_V <- 0.60

P_E_given_J <- 0.08
P_E_given_A <- 0.05
P_E_given_V <- 0.01
```

Overall percentage of error Using the Law of Total Probability:

$$P(E) = P(E \mid J)P(J) + P(E \mid A)P(A) + P(E \mid V)P(V)$$

```r
P_E <- P_E_given_J * P_J + P_E_given_A * P_A + P_E_given_V * P_V
P_E
```

```
## [1] 0.029
```

Answer: $P(E) = 0.029$, so the overall error rate is **2.9%**.

Most likely author of an erroneous program Using Bayes' Theorem for each employee:

$$P(J \mid E) = \frac{P(E \mid J)P(J)}{P(E)}, \quad P(A \mid E) = \frac{P(E \mid A)P(A)}{P(E)}, \quad P(V \mid E) = \frac{P(E \mid V)P(V)}{P(E)}$$

```r
P_J_given_E <- (P_E_given_J * P_J) / P_E
P_A_given_E <- (P_E_given_A * P_A) / P_E
P_V_given_E <- (P_E_given_V * P_V) / P_E

results <- data.frame(
  Employee = c("Jane", "Amy", "Ava"),
  `P(Employee | Error)` = c(P_J_given_E, P_A_given_E, P_V_given_E),
  check.names = FALSE
)
results
```

```
##   Employee P(Employee | Error)
## 1     Jane           0.2758621
## 2      Amy           0.5172414
## 3      Ava           0.2068966
```

Answer:

Employee	P(Employee | Error)
Jane	0.2759 (27.59%)
Amy	0.5172 (51.72%)
Ava	0.2069 (20.69%)

Amy is the most likely person to have written a program containing an error (51.72%). Even though Jane has the highest individual error rate (8%), Amy does far more of the total programming (30% vs. Jane's 10%), so in absolute terms more of the company's errors trace back to her work.